

Exact Global Optimality for Packing Ten Unequal Circles

The unequal-circle proof project

October 9, 2026

Abstract

We consider ten pairwise non-overlapping circles with radii $r_i = \sqrt{i}$, $1 \leq i \leq 10$, inside a circular container. We give a computer-assisted proof that the minimum container radius is an exact value R_{crit} , defined as the unique solution of two explicit angular equations in a rational rectangle. A matching ten-circle realization exists at this value, and a finite global certificate excludes every smaller radius. The lower-bound proof combines a complete radial and angular case tree, a two-inequality angle barrier for the main seven-circle order, and negative-cycle certificates for the four exceptional cases. For every hypothetical packing with $R < R_{\text{crit}}$, the composition accounts for all 14,757,888 sector assignments; the radial tree itself covers the larger domain $R \leq U$. The proof decisions in the replay use integer and rational arithmetic; the result is a reproducible computer-assisted proof, not an end-to-end theorem-prover formalization.

1 Introduction

Let $r_i = \sqrt{i}$ for $1 \leq i \leq 10$. The problem is to find the smallest radius R for which there are centers $p_1, \dots, p_{10} \in \mathbb{R}^2$ satisfying

$$\|p_i\| + r_i \leq R \quad (1 \leq i \leq 10), \quad (1)$$

$$\|p_i - p_j\| \geq r_i + r_j \quad (1 \leq i < j \leq 10). \quad (2)$$

Write R^* for this minimum. The minimum is attained: after fixing a bounded interval of candidate radii, all center coordinates lie in a compact set, and the constraints are closed.

This paper proves exact global optimality, rather than only a lower bound for a selected subfamily or a local statement near a proposed configuration. The proof is a composition of geometric inequalities with finite certificates. Its central idea is to turn each possible placement into one of finitely many radial boxes and cyclic-sector assignments. Most assignments close directly by rational radial or angular-cycle bounds. The remaining cases split into two geometric families:

1. a seven-circle core in one fixed cyclic order, ruled out by two angular sums whose monotonicities point in opposite directions as the radius of circle 10 varies;
2. four exceptional assignments, projected to a nine-circle order and ruled out by a finite negative-cycle certificate.

The angle barrier applies to every residual main-order case by a local radius estimate from the non-overlap and containment constraints. Finally, the same two angular equations that force the lower bound also define a configuration at R_{crit} .

We first define and realize the critical value, then reduce every possible packing below the tree cap to a finite radial and angular case list. The seven-circle barrier and the two exceptional-order certificates close the remaining cases. Section 8 describes the exact replay and its trust boundary.

Computer-assisted circle-packing research includes several related but distinct settings. Markót gives interval-based optimality proofs for 31, 32, and 33 equal circles in a square [1]. For unequal circles in circular containers, Huang et al. develop packing heuristics [2], while López and Beasley study a fixed-container formulation-space search heuristic [3]. Fekete, Keldenhich, and Scheffer prove a tight universal worst-case density bound of $1/2$ for arbitrary disk sets [4]. These works address equal-circle square packing, heuristic optimization, or universal density guarantees; here we determine the exact minimum radius for one prescribed ten-circle instance using a finite global certificate.

2 Problem and exact statement

2.1 Feasibility and the candidate value

For $\rho > 0$, call ρ feasible if there are centers satisfying (1)–(2) with $R = \rho$. Feasibility is monotone in ρ : a configuration feasible at ρ is feasible at every larger container radius. Thus a certified infeasibility result at one radius is a lower bound for R^* .

We define the exact candidate through the angular quantities introduced in Section 3. Let

$$\begin{aligned} J_t &= [1.0037160860750841, 1.0037160860750846], \\ J_R &= [8.3034681221114890, 8.3034681221114900]. \end{aligned}$$

The number R_{crit} is the R -coordinate of the unique pair $(t_{\text{crit}}, R_{\text{crit}}) \in J_t \times J_R$ satisfying

$$A(t_{\text{crit}}, R_{\text{crit}}) = 2\pi, \quad B(t_{\text{crit}}, R_{\text{crit}}) = 2\pi. \quad (3)$$

The existence and uniqueness of this pair, as well as the strict upper endpoint inequality, are proved by exact rational interval checks in Section 5. In particular,

$$8.303468122111489 \leq R_{\text{crit}} < 8.303468122111490 = U. \quad (4)$$

The equations, rather than a rounded decimal, define the exact value in this paper.

Theorem 2.1 (Global optimality). *For ten circles of radii $1, \sqrt{2}, \dots, \sqrt{10}$ in a circular container,*

$$\boxed{R^* = R_{\text{crit}}}.$$

More precisely, no packing exists for any $R < R_{\text{crit}}$, and a packing exists at $R = R_{\text{crit}}$.

The theorem determines the optimal radius; it does not assert uniqueness of the packing that attains it.

The exhaustive radial case tree covers every configuration with $R \leq U$. Its residual cases are closed by the angle barrier and the nine-circle negative-cycle certificate. Since $R_{\text{crit}} < U$, these facts cover every putative packing below R_{crit} . The matching construction is given in Section 3.3.

2.2 Exact specification and scope

The value in Theorem 2.1 is the exact root-defined real number, not the rounded decimal in (4); uniqueness refers to the defining root pair, not to the packing attaining the optimum.

3 Critical configuration and exact upper bound

3.1 The core order and two angular closure equations

Consider the seven-circle core

$$K = \{2, 5, 6, 7, 8, 9, 10\}, \quad W = K \setminus \{10\} = \{2, 5, 6, 7, 8, 9\}.$$

For a center p_i , put $s_i = \|p_i\|$. For any two circles with $r_i + r_j \leq s_i + s_j$ and $|s_i - s_j| < r_i + r_j$, define their necessary smaller central angle by

$$\phi_{ij}(s_i, s_j) = \arccos\left(\frac{s_i^2 + s_j^2 - (r_i + r_j)^2}{2s_i s_j}\right). \quad (5)$$

The law of cosines and non-overlap imply that the directed angular gap from one center direction to another is at least this smaller angle. In a cyclic order, all directed gaps sum to 2π .

The candidate core order is

$$(10, 5, 7, 9, 2, 8, 6).$$

For $i \in W$ put

$$a_i(R) = R - r_i, \quad D_i = r_{10} + r_i.$$

When the six circles in W are at the container wall, the angle between circle 10 at radial coordinate $t = s_{10}$ and circle i is

$$\alpha_i(t, R) = \arccos\left(\frac{t^2 + a_i(R)^2 - D_i^2}{2t a_i(R)}\right). \quad (6)$$

For two wall circles $i, j \in W$, their necessary angle is

$$\beta_{ij}(R) = 2 \arcsin \sqrt{\frac{r_i r_j}{a_i(R) a_j(R)}}. \quad (7)$$

Thus these are specializations of the general necessary angle:

$$\alpha_i(t, R) = \phi_{10,i}(t, a_i(R)), \quad \beta_{ij}(R) = \phi_{ij}(a_i(R), a_j(R)).$$

For the wall-wall pair, the cosine law gives $\cos \beta_{ij} = 1 - 2r_i r_j / (a_i a_j)$, yielding the half-angle form above. Define

$$C(R) = \beta_{79}(R) + \beta_{92}(R) + \beta_{28}(R) + \beta_{86}(R), \quad (8)$$

$$A(t, R) = \alpha_5(t, R) + \beta_{57}(R) + C(R) + \alpha_6(t, R), \quad (9)$$

$$B(t, R) = \alpha_7(t, R) + C(R) + \alpha_6(t, R). \quad (10)$$

The critical pair has a direct contact-and-closure description:

$$\alpha_5(t_{\text{crit}}, R_{\text{crit}}) + \beta_{57}(R_{\text{crit}}) = \alpha_7(t_{\text{crit}}, R_{\text{crit}}), \quad \alpha_7(t_{\text{crit}}, R_{\text{crit}}) + C(R_{\text{crit}}) + \alpha_6(t_{\text{crit}}, R_{\text{crit}}) = 2\pi. \quad (11)$$

The first equation matches the two contact routes from circle 10 to circle 7; the second closes the seven-circle cycle. Equivalently, with $F = A - B$ and $G = B - 2\pi$, the critical pair is the solution of $F = G = 0$.

For the stated order, the directed loop $10 \rightarrow 5 \rightarrow 7 \rightarrow 9 \rightarrow 2 \rightarrow 8 \rightarrow 6 \rightarrow 10$ has total angle 2π and therefore requires $A_{\text{radial}} \leq 2\pi$. A second loop $10 \rightarrow 7 \rightarrow 9 \rightarrow 2 \rightarrow 8 \rightarrow 6 \rightarrow 10$ skips circle 5 in the angle count, but circle 5 remains present in the packing. It gives the independent necessary inequality $B_{\text{radial}} \leq 2\pi$. These two inequalities are the key: their dependence on t has opposite signs near the critical pair.

3.2 The exact critical pair

For the global angular barrier, set

$$I = [0.8588, 1.1265].$$

This is the certified range of the radial coordinate $t = s_{10}$ and is distinct from the narrower root interval J_t in Section 2.

The following derivative formulas explain how the candidate is isolated. For $i \in \{5, 6, 7\}$ let

$$b_i = a_i(R)^2 - D_i^2, \quad H_i = 4a_i(R)^2 t^2 - (t^2 + b_i)^2.$$

They also satisfy

$$b_i = (a_i - D_i)(a_i + D_i) = (R + r_{10})(R - r_{10} - 2r_i), \quad (12)$$

$$H_i = ((a_i + D_i)^2 - t^2)(t^2 - (a_i - D_i)^2) \quad (13)$$

$$= ((R + r_{10})^2 - t^2)(t^2 - (R - r_{10} - 2r_i)^2). \quad (14)$$

On the geometric domain $0 < t < a_i + D_i$, the condition $H_i > 0$ is equivalent to the strict triangle inequalities $|a_i - D_i| < t < a_i + D_i$ for the three lengths t, a_i, D_i . Where the angles are in $(0, \pi)$,

$$\frac{\partial \alpha_i}{\partial t} = \frac{b_i - t^2}{t\sqrt{H_i}}, \quad \frac{\partial \alpha_i}{\partial R} = -\frac{a_i(R)^2 + D_i^2 - t^2}{a_i(R)\sqrt{H_i}}. \quad (15)$$

For a wall-wall angle,

$$\frac{\partial \beta_{ij}}{\partial R} = -\left(\frac{1}{a_i(R)} + \frac{1}{a_j(R)}\right) \tan \frac{\beta_{ij}(R)}{2}. \quad (16)$$

Here $\tan(\beta_{ij}/2) = \sqrt{r_i r_j / (a_i a_j - r_i r_j)}$. The sign certificates have different scopes. On the root rectangle $J_t \times J_R$, they prove $A_t(t, R) > 0$ and $B_t(t, R) < 0$, which gives uniqueness of the critical pair. For the global barrier, they also prove $A_t(t, R_{\text{crit}}) > 0$ and $B_t(t, R_{\text{crit}}) < 0$ for every $t \in I$. The radial derivatives satisfy $A_R(t, R) < 0$ and $B_R(t, R) < 0$ for every $t \in I$ and $R \in [7.9, 8.30346812212]$. The global case reduction supplies the stronger lower bound $R > 15947/2000$ for the main-order cases.

$$\begin{aligned} A_t(t, R) &> 0, & B_t(t, R) &< 0 & ((t, R) \in J_t \times J_R), \\ A_t(t, R_{\text{crit}}) &> 0, & B_t(t, R_{\text{crit}}) &< 0 & (t \in I), \\ A_R(t, R) &< 0, & B_R(t, R) &< 0 & (t \in I, 7.9 \leq R \leq 8.30346812212). \end{aligned} \quad (17)$$

For example, $A_t(t, R_{\text{crit}}) > 0$ follows from $(\alpha_5)_t(t, R_{\text{crit}}) > 0$ and $(\alpha_6)_t(t, R_{\text{crit}}) > 0$. To see the less immediate sign $B_t(t, R_{\text{crit}}) < 0$, abbreviate $a_i = a_i(R_{\text{crit}})$ and put $z = t^2$, $b_i = a_i^2 - D_i^2$, and $H_i = 4a_i^2 z - (z + b_i)^2$. Then

$$B_t(t, R_{\text{crit}}) = -\frac{z - b_7}{t\sqrt{H_7}} + \frac{b_6 - z}{t\sqrt{H_6}}.$$

Here $z - b_7 > 0$ and $b_6 - z > 0$ for $t \in I$. Squaring the positive terms shows that $B_t(t, R_{\text{crit}}) < 0$ follows from

$$\begin{aligned} (z - b_7)^2 H_6 - (b_6 - z)^2 H_7 &= 4zQ(z), \\ Q(z) &= D_6^2(z - b_7)^2 - D_7^2(z - b_6)^2. \end{aligned}$$

The identity follows by expansion using $b_i = a_i^2 - D_i^2$. Set $X = D_6(z - b_7)$ and $Y = D_7(b_6 - z)$. Both are positive on this interval, so

$$Q(z) = X^2 - Y^2 = (X - Y)(X + Y), \quad X > Y \iff z > \frac{D_6 b_7 + D_7 b_6}{D_6 + D_7}.$$

The exact interval checker verifies the coarser rational bounds

$$D_6 > 5.6, \quad D_7 < 5.9, \quad b_7 < -1.6, \quad 2.75 < b_6 < 2.8, \quad z > 0.73.$$

Consequently $X > 5.6(0.73 + 1.6) = 13.048$, while $Y < 5.9(2.8 - 0.73) = 12.213$. Thus $X > Y$, hence $Q > 0$ and $B_t < 0$.

To prove existence, use the functions $F = A - B$ and $G = B - 2\pi$ defined by the contact-and-closure equations above. The exact checker proves opposite signs for F on the lower and upper t -edges of the rational rectangle, and opposite signs for G on the lower and upper R -edges. Table 1 gives outward-rounded rational enclosures for the full edge ranges; these include variation along each edge rather than sampling only endpoints. Since $F_t > 0$, for every R in the rectangle there is exactly one $t(R)$ with $F = 0$. Continuity and the edge signs for G then give an R with $G = 0$ by the intermediate value theorem. This is precisely a solution of $A = B = 2\pi$.

Table 1: Outward rational enclosures for the four root-box edge signs.

Edge	Certified enclosure
$F(t_{\text{low}}, R), R \in J_R$	$[-4.68, -0.68] \times 10^{-16}$
$F(t_{\text{high}}, R), R \in J_R$	$[0.04, 0.44] \times 10^{-15}$
$G(t, R_{\text{low}}), t \in J_t$	$[1.75, 2.17] \times 10^{-16}$
$G(t, R_{\text{high}}), t \in J_t$	$[-2.47, -2.42] \times 10^{-15}$

The solution is unique. At fixed R , $A_t > 0$ gives at most one solution to $A = 2\pi$. If two simultaneous solutions had $R_1 < R_2$, then $A(t_1, R_2) < 2\pi$ and $B(t_1, R_2) < 2\pi$, because both decrease with R . Restoring $A = 2\pi$ at R_2 would require increasing t , while restoring $B = 2\pi$ would require decreasing t , since $A_t > 0$ and $B_t < 0$. Both cannot hold at the same t . Thus the pair is unique. The exact endpoint checks further show

$$t_{\text{crit}} \in J_t, \quad 8.303468122111489 \leq R_{\text{crit}} < 8.303468122111490.$$

All angle and derivative signs just used are checked without floating-point decisions. The verifier evaluates rational enclosures for square roots, uses Machin's identity with alternating arctangent series to enclose π , and uses half-angle arctangent expansions with rational remainder bounds for the inverse trigonometric terms.

3.3 An exact ten-circle witness

At $(t_{\text{crit}}, R_{\text{crit}})$, put circle 10 at radial distance t_{crit} and put each $i \in W$ at its wall radius $a_i(R_{\text{crit}}) = R_{\text{crit}} - r_i$. In the following display, $\alpha_i = \alpha_i(t_{\text{crit}}, R_{\text{crit}})$ and $\beta_{ij} = \beta_{ij}(R_{\text{crit}})$; the

polar angles are:

$$\begin{aligned}
\theta_{10} &= 0, \\
\theta_5 &= \alpha_5, \\
\theta_7 &= \alpha_5 + \beta_{57} = \alpha_7, \\
\theta_9 &= \theta_7 + \beta_{79}, \\
\theta_2 &= \theta_9 + \beta_{92}, \\
\theta_8 &= \theta_2 + \beta_{28}, \\
\theta_6 &= \theta_8 + \beta_{86} = 2\pi - \alpha_6.
\end{aligned}$$

The identities $\alpha_5 + \beta_{57} = \alpha_7$ and $\alpha_7 + \alpha_6 = 2\pi$ are exactly the two root equations. They make the eight pairs

$$(10, 5), (10, 7), (10, 6), (5, 7), (7, 9), (9, 2), (2, 8), (8, 6)$$

tangent. The remaining 13 pairs within the seven-circle core are strictly separated throughout the certified root rectangle.

Place the three other centers at the following fixed rational coordinates:

$$p_1 = (-3.9798, 6.1148), \quad p_3 = (6.4777, 0.8849), \quad p_4 = (5.5222, -2.7574).$$

(The terminating decimals here denote exact rationals.) The witness replay checks all 37 non-contact pairs among the ten circles and verifies strict container inclusion for circles 1, 3, 4 and circle 10, uniformly over the rational enclosure of $(t_{\text{crit}}, R_{\text{crit}})$. The six circles in W are on the wall, and all remaining inequalities have positive certified slack. Hence this is an actual feasible configuration at the exact root, not merely a rounded numerical picture.

Figure 1 shows the witness at rounded coordinates for orientation only. The proof uses the exact root-defined polar coordinates above and the rational auxiliary centers; it does not infer feasibility from the drawing.

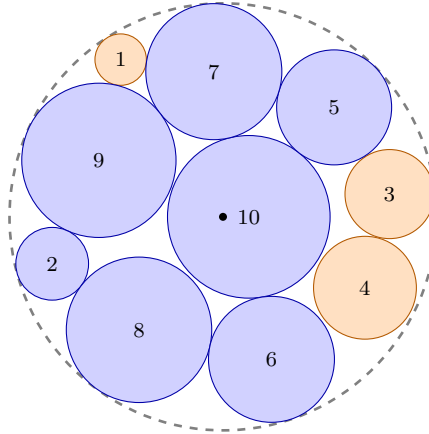
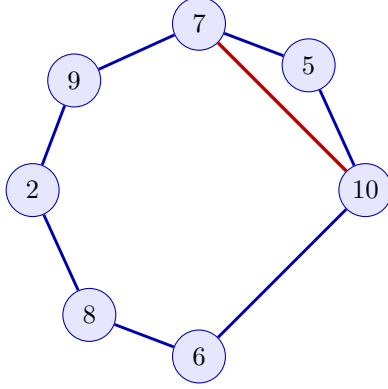


Figure 1: The ten-circle witness. Blue circles form the seven-circle core; orange circles are the three fixed rational auxiliary centers. The drawing uses rounded root coordinates and is illustrative only.

The eight active contacts within the core have a compact graph description. The seven edges of the cyclic chain follow the order used by the angular barrier, and the additional chord is the contact $(10, 7)$. The contact graph is shown in Figure 2.



cyclic contact chain additional chord (10,7)

Figure 2: Contact graph of the seven-circle core. The blue cycle contributes seven tangencies; the red chord is the eighth core tangency.

4 Global reduction to finite cases

A packing has ten radial coordinates $s_i = \|p_i\|$ and ten center directions. The proof infrastructure first subdivides the allowed radial domain by rational midpoint splits. On each radial box, exact necessary conditions determine which cyclic orders and angular-sector assignments remain possible. These assignments form the finite ledger called the global radial tree.

At a terminal case, one of the following happens:

- interval-safe radial inequalities contradict containment or non-overlap;
- exact angular constraints contradict the 2π budget, witnessed by a certified negative cycle in a difference-constraint graph;
- the assignment has the main seven-circle order and radial range, so the two-sum angle barrier applies;
- the assignment is one of the four alternative masks, so its nine-circle projection is covered by the negative-cycle tree.

Every split is rational and its children cover the parent box exactly. Thus the finite tree is a cover argument, not a sampling of candidate configurations.

The global cap is

$$U = 8.303468122111490.$$

Because the exact root lies strictly below this cap, it is enough for the tree to cover all configurations up to U ; no comparison of floating approximations is used to bridge a gap. This tree is a coverage certificate, not a claim that every radius $R \leq U$ is infeasible. The composed exclusion applies to hypothetical packings with $R < R_{\text{crit}}$; the exact witness in Section 3.3 establishes feasibility at $R = R_{\text{crit}}$.

The proof-flow diagram in Figure 3 previews how the finite tree closures and the two geometric barriers combine.

4.1 The four largest circles force a four-sector skeleton

The circles with labels 7, 8, 9, 10 provide the global angular scaffold. For every feasible packing with $R \leq U$, containment and pairwise non-overlap imply

$$s_i \leq U - r_i, \quad s_i \geq r_i + 2r_j - U \quad (i \neq j). \quad (18)$$

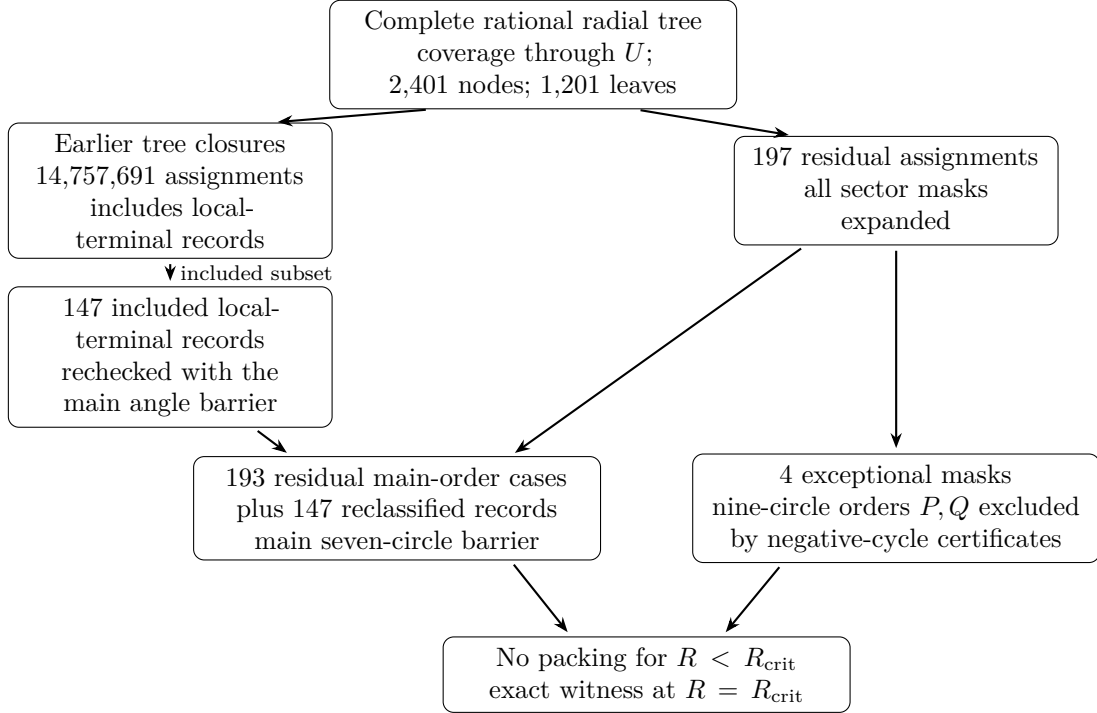


Figure 3: The finite lower-bound proof. The dashed arrow marks the 147 local-terminal records as a subset of the earlier closure count; they are rechecked through the main angle barrier and are not extra sector assignments. The case rules exclude hypothetical packings below R_{crit} , while the exact witness attains the boundary value.

The first inequality is containment. For the second, use $r_i + r_j \leq \|p_i - p_j\| \leq s_i + s_j$ and $s_j \leq U - r_j$. In particular, the directions of circles 7, 8, 9, 10 are defined: the exact rational enclosures used by the certificate give the safe lower bounds $s_7 > 0.6668$, $s_8 > 0.8495$, $s_9 > 1.0210$, and $s_{10} > 0.8588$.

Lemma 4.1 (The four largest centers are not contained in a semicircle). *For every packing with $R \leq U$, the four center directions of circles 7, 8, 9, 10 are not all contained in any closed semicircle.*

Proof. For $i, j \in \{7, 8, 9, 10\}$, the law of cosines and (18) give a lower bound q_{ij} on the smaller central angle between p_i and p_j . The six accepted rational angle ticks are listed in Table 2; their exact acceptance check uses the four-corner lemma below and a rational cosine Taylor lower bound. The verifier expands all $4! = 24$ possible linear orders of these directions along a putative semicircle. In every order, the sum of the three consecutive pair bounds exceeds $33019/10000 = 3.3019$, whereas the common rational upper bound $\pi_+^{\text{coarse}} = 3927/1250 = 3.1416$ satisfies $\pi < \pi_+^{\text{coarse}}$. The strict margin is $1603/10000$. Four directions lying in a closed semicircle would have a linear order whose three consecutive angular gaps sum to at most π , contradicting these lower bounds. $\square$

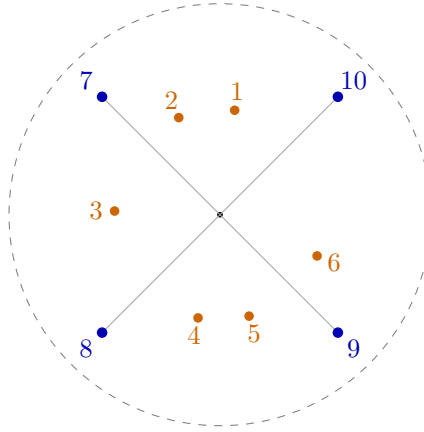
Consequently, when the four large directions are read counterclockwise, each of the four gaps between consecutive directions is strictly less than π : a gap at least π would put the other three directions in a closed semicircle. These four rays split the disk into four sectors. Assign each of circles 1, $\dots$, 6 to the sector containing its center direction; a direction exactly on a boundary may be assigned to either neighboring sector. If a small-circle center is the origin, assign it arbitrarily: the corresponding radial interval includes zero, every incident

Table 2: Certified lower bounds on the angles between the four largest circles. The minimum three-edge path sum over all 24 orders is 3.3019.

Pair	q_{ij}	Pair	q_{ij}
(7, 8)	1.0276	(7, 9)	1.0804
(7, 10)	1.1321	(8, 9)	1.1422
(8, 10)	1.1977	(9, 10)	1.2617

angular lower bound is set to zero, and the checker deletes that label only when testing projected orders. Thus this convention does not exclude a configuration. Figure 4 illustrates this assignment.

Rooting the cyclic order at circle 10 and identifying mirror images leaves exactly $(4 - 1)!/2 = 3$ skeleton classes. The verifier uses representatives $(10, 7, 8, 9)$, $(10, 7, 9, 8)$, and $(10, 8, 7, 9)$. The remaining six labels each choose one of four sectors, giving 4^6 assignments for every skeleton. The bit-mask encoding and the full count over all radial leaves are given in Section 7.



four large-circle rays define four sectors

Figure 4: Schematic sector assignment. Blue labels are the four large circles; each orange label is assigned to the sector containing its center direction. This drawing illustrates the combinatorial reduction and is not a packing or a metric certificate.

4.2 Radial boxes cover the full candidate domain

Let $\underline{r}_i$ be the rational lower endpoint for $\sqrt{i}$ stored in the certificate. The initial radial box is

$$s_i \in [\ell_i, u_i], \quad \ell_i = \max\left(0, \max_{j \neq i}(\underline{r}_i + 2\underline{r}_j - U)\right), \quad u_i = U - \underline{r}_i. \quad (19)$$

It contains every feasible radial vector with $R \leq U$. The certificate then recursively bisects one coordinate at a rational cut. The two closed children share the cut and cover the parent interval, so induction over the tree shows that its leaves cover the entire initial box; they are not samples of candidate placements. The replay checks all 2,401 nodes, all 1,200 splits, and all 1,201 leaves, including reachability and absence of duplicate nodes.

At a box with upper endpoints u_i , if $u_i + u_j < \underline{r}_i + \underline{r}_j$ for some pair, the box is empty by non-overlap. Otherwise the checker propagates the pair-sum inequalities to raise lower endpoints. This propagation is an outward-safe contraction: it removes only radial values that cannot satisfy $s_i + s_j \geq \underline{r}_i + \underline{r}_j$.

4.3 Exact angular relaxations and box-closing rules

For a pair with required center distance $d_{ij} = r_i + r_j$, define

$$c_{ij}(x, y) = \frac{x^2 + y^2 - d_{ij}^2}{2xy}.$$

By (5), the radial cosine is the cosine of the necessary angle:

$$c_{ij}(x, y) = \cos \phi_{ij}(x, y).$$

Thus that angle is $\phi_{ij}(x, y) = \arccos(c_{ij}(x, y))$. On a positive radial rectangle, c_{ij} attains its maximum at a corner:

Lemma 4.2 (Four-corner bound for the radial cosine). *Let $d > 0$ and let x, y range over a positive rectangle. The maximum of*

$$c(x, y) = \frac{x^2 + y^2 - d^2}{2xy}$$

on that rectangle is attained at one of its four corners.

Proof. For fixed $y > 0$,

$$\frac{\partial c}{\partial x} = \frac{x^2 - y^2 + d^2}{2yx^2}.$$

Its numerator is strictly increasing in x and has at most one zero; if a zero occurs, the derivative changes from negative to positive, so the stationary point is a minimum. The maximum over the x -interval is therefore at an endpoint. Applying the same argument in y to each of those two edges proves the corner claim. $\square$

The certificate chooses a rational angle tick $q_{ij} \geq 0$ for each pair. It is accepted only if a rational Taylor lower bound for $\cos(q_{ij})$ dominates the four-corner upper bound for c_{ij} . Since cosine decreases on $[0, \pi]$, every feasible smaller central angle is then at least q_{ij} . If a radial interval allows a center at the origin, its incident ticks are set to zero rather than assigning a direction that may not exist.

For the global-tree replay, the directed millionth enclosure is

$$\pi_-^{\text{tree}} = \frac{3141592}{10^6} < \pi < \pi_+^{\text{tree}} = \frac{3141593}{10^6}.$$

For a proposed cyclic order, resolve any equal-angle labels in an arbitrary order within each tied block, and lift the angles to a weakly increasing sequence $\theta_0 \leq \dots \leq \theta_{n-1} < \theta_0 + 2\pi$. A pair with equal directions has zero certified lower tick, so this tie-breaking does not change its difference constraints. Write i_a for the label at position a . If position a precedes position b , the necessary pair-angle bounds give

$$q_{i_a i_b} \leq \theta_b - \theta_a \leq 2\pi_+^{\text{tree}} - q_{i_a i_b}.$$

The four large-circle skeleton adds the fact that its four consecutive gaps are below π . These are difference constraints of the form $\theta_v - \theta_u \leq w(u, v)$.

Lemma 4.3 (Negative-cycle criterion for angular constraints). *If a finite system of difference constraints $\theta_v - \theta_u \leq w(u, v)$ contains a directed cycle whose edge weights sum to a strictly negative number, the system has no solution.*

Proof. Summing the inequalities around the cycle makes the left side telescope to zero, so feasibility would imply $0 \leq \sum w(u, v) < 0$. The checker also verifies that each recorded cycle is one simple directed cycle and recomputes its sum with exact rational arithmetic. $\square$

Some boxes yield a feasible potential vector for these relaxed angular constraints. The verifier labels such a vector **ANGLE_MODEL**; it is only a model of necessary inequalities, not a set of Cartesian centers and not a packing witness. Such a box remains open for the later geometric theorems.

In a smaller number of cases, the angular constraints imply a rational upper bound $\Gamma_{ij} < \pi$ on the smaller angle between a pair. If $c^- \leq \cos(\Gamma_{ij})$ is the rational Taylor lower bound, then the actual smaller angle γ_{ij} satisfies $\cos(\gamma_{ij}) \geq c^-$. For each disk k , write $I_k = [\ell_k, u_k]$ for its radial interval. Throughout the box this gives

$$\|p_i - p_j\|^2 \leq \max_{x \in I_i, y \in I_j} (x^2 + y^2 - 2xyc^-).$$

The quadratic is convex on the rectangle because $-1 \leq c^- \leq 1$, so its maximum is attained at a corner. If its verified upper bound is below $(r_i + r_j)^2$, the circles must overlap, closing the box by a **PAIR_DISTANCE** witness. Thus the original tree uses radial pair-sum contradictions, angular negative cycles, and these angle-to-distance contradictions. The retained **ANGLE_MODEL** cases and the 147 terminal records formerly closed by a local rule are checked again by the global geometric barriers; the local conclusion is not used as a final case closure.

5 The seven-circle angular barrier

5.1 Why the radial coordinates can be replaced by wall values

The main order cases do not assume that the circles $i \in W$ are at the wall. This is important: the theorem does not move disks, and it does not assume the older sufficient conditions for physically pushing disks outward.

For $d_{ij} = r_i + r_j$, differentiation of (5) gives

$$\frac{\partial \phi_{ij}}{\partial s_i} = -\frac{s_i^2 - s_j^2 + d_{ij}^2}{2s_i^2 s_j \sin \phi_{ij}}. \quad (20)$$

The case boxes imply, for $t = s_{10} \in I = [0.8588, 1.1265]$ and $\frac{15947}{2000} \leq R \leq U$,

$$r_{10} + r_i - t \leq s_i \leq U - r_i \quad (i \in W).$$

The left inequality follows from $r_{10} + r_i \leq \|p_{10} - p_i\| \leq s_{10} + s_i$; the right one is containment. The actual verifier rounds the square-root endpoints outward to rational bounds before doing its checks.

For most coordinates, (20) has a directly certified negative sign in each angular sum. Three shared coordinates require a combined derivative check because their two adjacent pair contributions can have opposite signs. For $(i, g, b) = (9, 7, 2), (8, 6, 2), (2, 9, 8)$, exact rational bounds are applied to the combined numerator

$$\Lambda_i = \frac{X_{ig}}{s_g \sin \phi_{ig}} + \frac{X_{ib}}{s_b \sin \phi_{ib}}, \quad X_{ij} = s_i^2 - s_j^2 + (r_i + r_j)^2.$$

The derivative of the two adjacent angles with respect to s_i is $-\Lambda_i/(2s_i^2)$. Exact lower bounds for Λ_i are:

coordinate s_i	certified lower bound for Λ_i
s_9	> 3.193809
s_8	> 0.143756
s_2	> 0.589640

The smaller angular sines in the two potentially adverse pairs (9, 2) and (2, 8) are bounded below by $3/5$; the remaining terms have positive rational lower bounds. For the individual terms, the checker certifies $X_{ij} > 0$ on the same radial rectangles. For every adjacent pair, the initial feasible packing satisfies $s_i + s_j \geq r_i + r_j$ by non-overlap and the triangle inequality. Each substitution increases a radial coordinate, so this weak inequality persists throughout the interpolation. The exact box check separately proves $|s_i - s_j| < r_i + r_j$ throughout. Thus the angle remains defined in $[0, \pi)$; when $s_i + s_j = r_i + r_j$ it equals 0, and the monotonicity conclusion extends to that endpoint by continuity from the interior. Both full radial sums are therefore non-increasing in every s_i .

Starting from any feasible packing in the main order, replace its six radial coordinates s_i one at a time by $R - r_i$ only inside the angle formulas. This is an analytic comparison, not a physical movement. Since each angular sum is non-increasing as any s_i increases, the wall-value sums satisfy

$$A(t, R) \leq A_{\text{radial}} \leq 2\pi, \quad B(t, R) \leq B_{\text{radial}} \leq 2\pi.$$

This is the wall-free angular barrier.

Lemma 5.1 (A radius floor for the main core). *If a feasible main-core packing has $t = s_{10} \leq 1.1265$, then*

$$R > \frac{15947}{2000} = 7.9735.$$

Proof. Non-overlap of circles 9 and 10 gives $s_9 + t \geq r_9 + r_{10} = 3 + \sqrt{10}$. Containment of circle 9 gives $R \geq 3 + s_9$. Since $\sqrt{10} > 3.1$ and $t \leq 1.1265$,

$$R \geq 6 + \sqrt{10} - t \geq 6 + \sqrt{10} - 1.1265 > 6 + 3.1 - 1.1265 = 7.9735.$$

□

5.2 Excluding every radius below the critical value

Suppose a main-order packing occurs in a tree box. Its disk-10 coordinate lies in I , so Lemma 5.1 puts its radius in the certified derivative range. If $R < R_{\text{crit}}$, then by (5.1), it must satisfy $A(t, R) \leq 2\pi$ and $B(t, R) \leq 2\pi$.

If $t \geq t_{\text{crit}}$, then $A_R < 0$ on the interval from R to R_{crit} , while $A_t > 0$ at R_{crit} , so

$$A(t, R) > A(t, R_{\text{crit}}) \geq A(t_{\text{crit}}, R_{\text{crit}}) = 2\pi,$$

a contradiction. If $t \leq t_{\text{crit}}$, then $B_t < 0$ and $B_R < 0$ imply

$$B(t, R) > B(t, R_{\text{crit}}) \geq B(t_{\text{crit}}, R_{\text{crit}}) = 2\pi,$$

also a contradiction. These two cases cover every value of t . Thus any main-order case satisfying the tree's radial hypotheses is impossible below R_{crit} .

6 The four exceptional cases: a negative-cycle proof

The four remaining assignments have a different large-circle skeleton and are not covered by the preceding seven-circle theorem. Their masks are $(5, 16, 40, 2)$ (three assignments) and $(4, 16, 41, 2)$ (one assignment). After projecting away circle 1, their nine-circle order is

$$P = (10, 3, 7, 5, 8, 4, 6, 9, 2).$$

The only ambiguity in the relative placement of circles 4 and 6 gives the second possible order

$$Q = (10, 3, 7, 5, 8, 6, 4, 9, 2).$$

The exact certificate covers both orders; the four assignments in the global tree project to P .

6.1 From radial boxes to angle-difference constraints

Fix either order. For a radial box $s_i \in [\ell_i, u_i]$, containment gives $0 \leq s_i \leq U_+ - r_i$, where

$$U_+ = 8.30346812212 > U.$$

Non-overlap also requires $s_i + s_j \geq r_i + r_j$. This inequality safely tightens the lower endpoints of each box. If any lower endpoint is zero, the angle lower bound involving that coordinate is set to zero, which is conservative.

For each pair, a nonnegative rational angle tick q_{ij} is stored. It is accepted only after an exact rational Taylor inequality certifies $\cos(q_{ij}) > c_{ij}$ throughout the box. Floating-point inverse cosine is used only when searching for candidate ticks; it has no role in replay. Consequently any feasible angular difference Δ_{ij} in the fixed cyclic order must satisfy

$$q_{ij} \leq \Delta_{ij} \leq 2\pi_+^{\text{coarse}} - q_{ij},$$

where $\pi_+^{\text{coarse}} = 3927/1250 = 3.1416$ is the same rational upper bound used in the four-large-circle lemma. The nine-circle replay verifies $\pi < \pi_+^{\text{coarse}}$ by an exact Machin-series bound.

The four-corner bound and negative-cycle criterion from Section 4.3 apply to these ticks. The certificate replays every listed cycle with its exact edge weights; Table 3 summarizes the two trees.

6.2 Finite certificate

The initial radial domain is covered by rational boxes. Each unresolved box is bisected at a rational midpoint; the verifier checks that the children cover the parent and that no box is duplicated or omitted. Every leaf has a recorded negative cycle whose edge weights are recomputed from the radial box and rational angle ticks.

Table 3: Exact replay of the two nine-circle order certificates.

Order	Nodes	Splits	Negative-cycle leaves	Unresolved
P	231	115	116	0
Q	77	38	39	0
Total	308	153	155	0

In Table 4, each displayed terminating decimal for q is the exact rational tick $k/100000$. For a concrete leaf of order P , the recorded simple cycle is $4 \rightarrow 8 \rightarrow 5 \rightarrow 7 \rightarrow 3 \rightarrow 10 \rightarrow$

$9 \rightarrow 6 \rightarrow 4$. Its angle ticks and edge weights are shown in Table 4. The edge $10 \rightarrow 9$ is the unique edge oriented forward in the chosen linear lift and has weight $2\pi_+^{\text{coarse}} - q$; the other edges have weight $-q$. Hence the exact cycle sum is

$$2\pi_+^{\text{coarse}} - \frac{354613}{50000} = -\frac{40453}{50000} = -0.80906 < 0,$$

using the shared rational upper bound. This is one of the 116 leaves in the P tree, not a numerical approximation to an angular cycle.

Table 4: An explicit negative cycle from the first P -tree leaf.

Edge	q	Weight	Edge	q	Weight
$4 \rightarrow 8$	1.02792	$-q$	$8 \rightarrow 5$	0.90299	$-q$
$5 \rightarrow 7$	0.85614	$-q$	$7 \rightarrow 3$	0.36964	$-q$
$3 \rightarrow 10$	1.16698	$-q$	$10 \rightarrow 9$	1.26172	$2\pi_+^{\text{coarse}} - q$
$9 \rightarrow 6$	1.01613	$-q$	$6 \rightarrow 4$	0.49074	$-q$

Thus both projected orders are excluded for every $R \leq U_+$, including the four global-tree assignments because $U < U_+$.

7 Composition of the global finite proof

The proof uses a complete rational-split tree for the radial domain up to U . This tree covers the candidate radial domain; it does not by itself claim infeasibility at every radius below U . Its replay checks

$$2,401 \text{ nodes}, \quad 1,200 \text{ splits}, \quad 1,201 \text{ leaves}.$$

7.1 Sector masks and complete case accounting

For each radial leaf, the checker considers the three canonical cyclic orders of the four-circle skeleton $\sigma = (\sigma_0, \sigma_1, \sigma_2, \sigma_3)$, rooted at circle 10 and identified up to reflection. The six remaining circles have labels $1, \dots, 6$. Each is assigned to one of the four sectors after σ_k and before σ_{k+1} (indices modulo four). If S_k is the set assigned to sector k , its integer mask is

$$M_k = \sum_{i \in S_k} 2^{i-1}, \quad k = 0, 1, 2, 3. \quad (21)$$

The tuple (M_0, M_1, M_2, M_3) records the assignment: every label occurs in exactly one sector, and each of the six labels has four choices. Thus there are 4^6 masks for each skeleton and $3 \cdot 4^6 = 12,288$ sector assignments per radial leaf. With 1,201 leaves, the total is

$$1,201 \cdot 3 \cdot 4^6 = 14,757,888.$$

For example, the mask tuple $(5, 16, 40, 2)$ assigns label sets $(\{1, 3\}, \{5\}, \{4, 6\}, \{2\})$ to the four successive sectors. When all angular lower bounds incident to a disk vanish, the replay may clear that label's bit in an effective mask; this merges equivalent order checks but does not remove any of the $3 \cdot 4^6$ original sector assignments from the coverage count.

Under the tree's original rules, 14,757,691 assignments were closed and 197 remained unresolved. The closed count includes local-terminal cases. The 147 local records described below are rechecked because their earlier closure used a local premise. The residual assignments divide as follows:

- The 193 main cases have seven-circle core order $(10, 5, 7, 9, 2, 8, 6)$ and meet the wall-free barrier hypotheses;
- Four exceptional masks have nine-circle projection P after deleting circle 1. Both P and Q are covered by the negative-cycle certificate.

The 193 residual main-order cases have disk-10 radial intervals contained in $[0.8588, 1.1265]$, which places them in the range required by the seven-circle barrier. Both exceptional orders P and Q have seven-circle core projection $(10, 7, 5, 8, 6, 9, 2)$, so they are not passed to the main-order barrier. The masks are $(5, 16, 40, 2)$ in three cases and $(4, 16, 41, 2)$ in one. The composition verifier also reclassifies 145 core-order terminal models and 2 full-order terminal witnesses through the same seven-circle angle barrier. This removes their dependence on the previous local terminal rule. Each of these 147 records has the main core order and the required disk-10 interval, so Lemma 5.1 supplies the radius range required by the barrier. These are reclassifications of terminal records already included in the tree’s prior count, not additional sector assignments. For every hypothetical packing with $R < R_{\text{crit}}$, the composed rules rule out all 14,757,888 assignments. This statement does not exclude the feasible boundary case $R = R_{\text{crit}}$.

Lemma 7.1 (Completeness of sector-order projection). *Fix a four-circle skeleton σ in cyclic order and effective sector masks $M = (M_0, M_1, M_2, M_3)$ for the remaining circles with positive angular lower bounds. The set $\Pi(\sigma, M)$ formed by preserving the skeleton order and taking every permutation of the assigned labels within each sector contains every order compatible with that assignment. Every full order, after deleting labels whose angular lower bounds vanish, restricts to one of these projections. Consequently, if every order in $\Pi(\sigma, M)$ has its seven-circle restriction equal to $(10, 5, 7, 9, 2, 8, 6)$, the main angle barrier applies to every packing in that mask. If every order instead has its nine-circle restriction, after deleting circle 1, in $\{P, Q\}$, the negative-cycle certificate applies to every packing in that mask.*

Proof. Read any compatible cyclic order starting at the first skeleton circle. Each remaining circle with a positive angular lower bound lies in one sector between consecutive skeleton circles. Its within-sector order is one of the permutations enumerated in $\Pi(\sigma, M)$; ties can be assigned either consistent order and both are included. Circles with zero lower bounds are deleted, which only relaxes the necessary inequalities. The composition checker constructs the whole projection set for each residual mask and verifies the required restrictions for every member; it does not infer them from the single order stored with an angular model. $\square$

For the 197 residual sector assignments, the replay checks 790 projected orders in total. All 772 projections in the main-order groups restrict to the stated seven-circle order. All 18 projections in the exceptional groups restrict, after deleting circle 1, to P or Q . The 147 local terminal models are separately replayed order by order through the main barrier. These finite checks make the connection from the case tree to the two geometric theorems explicit.

Proof of Theorem 2.1. Let $R < R_{\text{crit}}$. Since $R_{\text{crit}} < U$, every putative packing belongs to the certified radial-tree domain. Every sector assignment is closed either by the original exact radial/angular checks, by the main-order barrier, by the alternative nine-circle certificate, or by the independent reclassification of the 147 local terminals. Thus no packing exists below R_{crit} . Section 3.3 constructs a packing at R_{crit} , so $R^* = R_{\text{crit}}$. $\square$

8 Reproducibility and trust boundary

The complete composition is replayed from the repository root with

```
uv run -locked python \
tools/verify_global_optimality_completion.py
```

This replay checks the radial tree, theorem links, exact root and witness, and both exceptional orders. The JSON report is written to `artifacts/`.

All decisive arithmetic in these replays is integer or rational arithmetic. Square roots are enclosed with integer square-root bounds; the trigonometric inequalities are reduced to rational Taylor bounds and rational enclosures of π . Floating-point routines may help generate candidate split points or angle ticks, but the verifier rechecks the accepted claims using exact inequalities. This separates fast certificate discovery from the smaller checking computation.

The trust boundary is explicit. This is a computer-assisted proof whose soundness depends on the correctness of the Python verifiers, the Python runtime’s integer and `Fraction` semantics, and the certificate inputs whose hashes the composition checker records. The proof is not formalized end-to-end in Lean. In particular, references to old Lean developments or the MPFI generator do not substitute for the exact replay used here.

The composition entry point rejects Python’s optimized modes, because some component checkers use `assert` statements for certificate conditions; running it with `-O` or `-OO` would otherwise remove those checks. The SHA-256 manifest covers the exact case tree, theorem data, verifiers, locked Python environment, composition note, root README, and the manuscript source and PDF. All of these are committed as one reproducible source unit.

The composition also replays the previous local-terminal classifier to identify the 145 core-order models and two full-order records that had been closed by that rule. It does not accept those local conclusions as final closures: each selected record is separately checked against the seven-circle angle barrier and its required radial interval. The local verifier therefore remains in the case-selection and replay dependency chain, while the final geometric implication for those records is the global barrier.

The repository’s global-optimality completion note and the `proof/global_completion/` directory contain the individual theorem proofs and certificate details. The composition report records SHA-256 hashes for these inputs so a replay can be tied to the checked inputs.

9 Conclusion

The finite tree and its geometric terminal theorems give a global lower bound at the exact angle-defined value R_{crit} . The root construction supplies a ten-circle packing at the same value. Therefore the minimum container radius is exactly R_{crit} , as claimed in Theorem 2.1.

The coupled-angle equations give an exact specification of the optimum; the finite decimals serve only as certified enclosures for the root and as convenient data for replay.

References

- [1] Mihály Csaba Markót, Improved interval methods for solving circle packing problems in the unit square, *Journal of Global Optimization* 81(3) (2021), 773–803. [doi:10.1007/s10898-021-01086-z](https://doi.org/10.1007/s10898-021-01086-z).
- [2] Wenqi Huang, Yu Li, Chumin Li, and Ruchu Xu, New heuristics for packing unequal circles into a circular container, *Computers & Operations Research* 33(8) (2006), 2125–2142. [doi:10.1016/j.cor.2005.01.003](https://doi.org/10.1016/j.cor.2005.01.003).

- [3] C. O. López and J. E. Beasley, A formulation space search heuristic for packing unequal circles in a fixed size circular container, *European Journal of Operational Research* 251(1) (2016), 64–73. [doi:10.1016/j.ejor.2015.10.062](https://doi.org/10.1016/j.ejor.2015.10.062).
- [4] S. P. Fekete, P. Keldenich, and C. Scheffer, Packing disks into disks with optimal worst-case density, *Discrete & Computational Geometry* 69 (2023), 51–90. [doi:10.1007/s00454-022-00422-8](https://doi.org/10.1007/s00454-022-00422-8).